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1-1 PRIVATE TUTORS

ALGEBRA

Solving Equations

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KEY WORDS equation *(whārite)* · solve *(whakaoti)* · solution *(otinga)* · inverse operation *(mahi kōaro)* · both sides

A One step: adding and subtracting ●○○

Remember: Undo what has been done to x, and do the same to both sides. To undo $+7$, subtract 7. To undo -3, add 3.

1. Solve.

a) $x + 7 = 12 \quad x =$ _____ **b)** $x - 3 = 6 \quad x =$ _____ **c)** $x + 12 = 20 \quad x =$ _____

d) $x - 6 = -2 \quad x =$ _____ **e)** $x + 9 = 30 \quad x =$ _____ **f)** $x - 15 = -5 \quad x =$ _____

g) $x + 25 = 55 \quad x =$ _____ **h)** $x - 8 = -8 \quad x =$ _____ **i)** $x + 4 = 4 \quad x =$ _____

j) $x - 20 = 15 \quad x =$ _____ **k)** $x + 18 = 25 \quad x =$ _____ **l)** $x - 11 = 0 \quad x =$ _____

B One step: multiplying and dividing ●●○

Remember: To undo $\times 4$, divide by 4. To undo $\div 3$, multiply by 3. Remember that $4x$ means $4 \times x$.

2. Solve.

a) $4x = 24 \quad x =$ ___ **b)** $\frac{x}{3} = 4 \quad x =$ ___ **c)** $5x = 45 \quad x =$ ___ **d)** $\frac{x}{2} = 7 \quad x =$ ___

e) $7x = 56 \quad x =$ ___ **f)** $\frac{x}{5} = 6 \quad x =$ ___ **g)** $10x = 120 \quad x =$ ___ **h)** $\frac{x}{4} = 9 \quad x =$ ___

i) $3x = 45 \quad x =$ ___ **j)** $\frac{x}{6} = 7 \quad x =$ ___ **k)** $9x = 0 \quad x =$ ___ **l)** $\frac{x}{10} = 7 \quad x =$ ___

C Two-step equations ●●●

Remember: Undo the adding or subtracting first, then undo the multiplying or dividing.

3. Solve. Show each step.

a) $2x + 3 = 11$ **b)** $3x - 5 = 13$ **c)** $5x + 1 = 36$ **d)** $4x + 7 = 15$

e) $6x - 4 = 26$ **f)** $2x + 9 = 15$ **g)** $10x - 7 = 13$ **h)** $3x + 8 = 8$

i) $\frac{x}{3} + 2 = 6$ **j)** $\frac{x}{2} - 1 = 4$ **k)** $7x + 4 = 25$ **l)** $8x - 9 = 23$

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D Checking a solution ●●○

> **Remember:** Substitute the value into the equation. If both sides come to the same number, it is a solution.

4. Is the value of x a solution of the equation? Show your working.

a) $x = 6$ and $x + 9 = 15$

b) $x = 7$ and $4x = 30$

c) $x = 3$ and $2x + 5 = 11$

d) $x = 5$ and $3x - 4 = 10$

e) $x = 10$ and $\frac{x}{2} + 3 = 8$

f) $x = 4$ and $5x + 1 = 21$

E Equations from words ●●●

5. Write an equation for each problem, then solve it.

a) I think of a number and add 17. The answer is 42. What is the number?

b) 6 identical books cost $54 altogether. How much does one book cost?

c) I think of a number, multiply it by 3 and add 4. The answer is 19. What is the number?

d) A taxi charges $4 plus $3 for each kilometre. A trip costs $25. How many kilometres long is the trip?

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6. Solve.

a) $4x + 6 = 16$

b) $15 - 2x = 7$

c) $3(x + 2) = 21$

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ALGEBRA

Solving Equations

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KEY WORDS equation *(whārite)* · solve *(whakaoti)* · solution *(otinga)* · inverse operation *(mahi kōaro)* · both sides

A One step: adding and subtracting ●○○

Remember: Undo what has been done to x, and do the same to both sides. To undo $+7$, subtract 7. To undo -3, add 3.

1. Solve.

a) $x + 6 = 14 \quad x =$ _____ **b)** $x - 4 = 3 \quad x =$ _____ **c)** $x + 14 = 23 \quad x =$ _____

d) $x - 5 = 1 \quad x =$ _____ **e)** $x + 8 = 31 \quad x =$ _____ **f)** $x - 12 = -2 \quad x =$ _____

g) $x + 35 = 55 \quad x =$ _____ **h)** $x - 9 = -9 \quad x =$ _____ **i)** $x + 6 = 6 \quad x =$ _____

j) $x - 30 = -5 \quad x =$ _____ **k)** $x + 16 = 25 \quad x =$ _____ **l)** $x - 13 = 0 \quad x =$ _____

B One step: multiplying and dividing ●●○

Remember: To undo $\times 4$, divide by 4. To undo $\div 3$, multiply by 3. Remember that $4x$ means $4 \times x$.

2. Solve.

a) $3x = 27 \quad x =$ ___ **b)** $\frac{x}{4} = 5 \quad x =$ ___ **c)** $6x = 42 \quad x =$ ___ **d)** $\frac{x}{2} = 9 \quad x =$ ___

e) $8x = 72 \quad x =$ ___ **f)** $\frac{x}{5} = 9 \quad x =$ ___ **g)** $10x = 150 \quad x =$ ___ **h)** $\frac{x}{3} = 11 \quad x =$ ___

i) $4x = 48 \quad x =$ ___ **j)** $\frac{x}{7} = 8 \quad x =$ ___ **k)** $6x = 0 \quad x =$ ___ **l)** $\frac{x}{10} = 9 \quad x =$ ___

C Two-step equations ●●●

Remember: Undo the adding or subtracting first, then undo the multiplying or dividing.

3. Solve. Show each step.

a) $2x + 5 = 11$ **b)** $3x - 4 = 11$ **c)** $4x + 1 = 25$ **d)** $5x + 8 = 18$

e) $6x - 5 = 19$ **f)** $2x + 7 = 15$ **g)** $10x - 9 = 21$ **h)** $4x + 9 = 9$

i) $\frac{x}{4} + 3 = 5$ **j)** $\frac{x}{2} - 3 = 4$ **k)** $7x + 5 = 19$ **l)** $9x - 8 = 19$

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D Checking a solution ●●○

> **Remember:** Substitute the value into the equation. If both sides come to the same number, it is a solution.

4. Is the value of x a solution of the equation? Show your working.

a) $x = 9$ and $x + 8 = 17$

b) $x = 9$ and $5x = 40$

c) $x = 4$ and $2x + 7 = 15$

d) $x = 6$ and $3x - 5 = 12$

e) $x = 12$ and $\frac{x}{3} + 2 = 6$

f) $x = 3$ and $5x + 2 = 17$

E Equations from words ●●●

5. Write an equation for each problem, then solve it.

a) I think of a number and add 19. The answer is 45. What is the number?

b) 7 identical books cost $63 altogether. How much does one book cost?

c) I think of a number, multiply it by 4 and add 3. The answer is 27. What is the number?

d) A taxi charges $5 plus $2 for each kilometre. A trip costs $23. How many kilometres long is the trip?

★ Extension

6. Solve.

a) $4x + 2 = 8$

b) $17 - 3x = 5$

c) $2(x + 4) = 14$

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